8 Features of Manufacturing Industries

8.3 – Characteristics, processes, application, advantages and disadvantages and the importance of considering accuracy of production and efficiency of modern manufacturing methods and systems when designing for manufacture for small, medium, and large-scale production:

e. Lean Manufacturing using Just-in-Time (JIT) systems

<u>Definition:</u> Lean Manufacturing

Just-in-Time (JIT) Manufacture

Lean Manufacturing
The *<u>five</u>* key stages of Lean Manufacturing

Key Stage		Content
1	**Value**	Focus on value in the context of what the customer / end user is prepared to pay for
2	**Value Stream**	Identify how ***value-adding*** and ***non-value-adding activities*** affect efficiency throughout production including: ▪ *For value-adding activities – i.e. machining, processing, painting, assembling etc* ▪ *For non-value-adding activities – i.e. scrapping, sorting, storing, counting, moving etc*
3	**Flow**	Design processes that result in ***uninterrupted flow*** – from raw material to delivery of finished product
4	**Pull**	Design manufacturing for ***'pull'*** of product through the process as a response to demand, *<u>rather than</u>* ***'push'*** of raw materials into the process by producing components irrespective of whether they are needed or not
5	**Perfection (Kaizen)**	Adopt an approach that ***continually improves*** working processes

1. Pull Tools (Kanban)

2. Perfection Tools (Kaizen)

3. Flexible Manufacturing Cells